Across the Channel: Social Networks and the Cross-Border Economic Effects of Brexit

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Abstract

French départements with stronger social connections to the United Kingdom—measured by Facebook's Social Connectedness Index—experienced housing price *appreciation* after the 2016 Brexit referendum ($\beta = 0.025$, $p = 0.031$). The effect concentrates in houses ($\beta = 0.035$), not apartments, and in Channel-facing départements; traditional UK expat areas show price *declines*. These patterns point to demand reallocation from UK buyers, not generic capital flows. Replacing the binary post-referendum indicator with the GBP/EUR exchange rate confirms a sterling depreciation channel ($p = 0.022$). However, baseline département price levels absorb much of the effect, and a German placebo is significant ($\beta = 0.045$, $p = 0.008$), indicating that internationally connected—and expensive—départements appreciated differentially regardless of UK-specific exposure. The UK-specific component is real but modest, operating within a broader post-2016 divergence of cosmopolitan and provincial France.

JEL Codes: F15, R31, L14, D85

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1. Introduction

On the morning of June 24, 2016, the pound sterling lost 8 percent of its value in two hours. The immediate question was what Brexit would cost the United Kingdom. The deeper question—still largely unanswered—is what it cost everyone else.

The sign of the answer is surprising. French départements with stronger social connections to the UK experienced housing price *appreciation* after the referendum, not decline. A one standard deviation increase in log SCI-weighted UK exposure predicts 0.7 percent higher prices per square meter ($\beta = 0.025$, $p = 0.031$), an effect that emerges at the referendum date and persists through the sample period. Social networks transmitted demand reallocation, not economic distress, across the Channel.

Three patterns sharpen the mechanism. First, effects concentrate entirely in houses ($\beta = 0.035$, $p = 0.011$), with null effects on apartments ($\beta = 0.006$, $p = 0.755$)—consistent with UK residential buyers, not institutional capital flows. Second, Channel-facing départements show effects five times the baseline, while traditional UK expat areas (Dordogne, Charente, Creuse) show price *declines*—suggesting demand redistribution within France rather than a uniform increase. Third, replacing the binary post-referendum indicator with the time-varying GBP/EUR exchange rate confirms a sterling depreciation channel ($p = 0.022$): when the pound weakened, UK-connected French départements appreciated more.

The identification exploits within-France variation in social connectivity, measured by Facebook's Social Connectedness Index (Bailey et al., 2018, 2021). Creuse and Charente—départements with established British expatriate communities—are far more connected to the UK than interior départements like Cantal or Nièvre. The empirical specification interacts log SCI with a post-referendum indicator, controlling for département and quarter-year fixed effects.

The main identification challenge is that UK-connected départements are also France's most cosmopolitan hubs. A German placebo—constructed identically using France–Germany SCI—is positive and significant ($\beta = 0.045$, $p = 0.008$), larger than the UK treatment effect. Controlling for baseline département price levels absorbs most of the UK coefficient ($\beta = 0.003$ with price controls, versus 0.025 without), revealing that much of the variation loads on the fact that expensive, internationally connected départements appreciated faster after 2016. Département-specific linear trends attenuate the UK effect to insignificance ($\beta = 0.015$, $p = 0.217$). The honest interpretation: the UK SCI coefficient captures a composite of UK-specific demand reallocation and a broader divergence between cosmopolitan and provincial France. Permutation inference ($p = 0.038$), the house/apartment heterogeneity, and the Channel-facing geographic concentration support a genuine UK-specific component—but it

operates within, and cannot be cleanly separated from, a larger European integration trend.

This paper contributes to three literatures. First, it extends the economics of social networks (Rauch, 2001; Chaney, 2014; Bailey et al., 2018, 2021) to cross-border transmission of political-economic shocks. Burchardi and Hassan (2013) showed that West German regions with social ties to East Germany experienced higher income growth after reunification; I apply the same logic across national borders and find that the sign of cross-border spillovers can be positive when the originating shock redirects demand. Second, the paper adds to the Brexit literature (Sampson, 2017; Born et al., 2019; Bloom et al., 2025; Hassan et al., 2024) by documenting positive cross-border spillovers—demand reallocation rather than contagion—to a neighboring country's housing markets. Third, the paper illustrates both the promise and limits of the SCI as an identification tool in cross-border settings: the index captures genuine social network variation, but this variation correlates with baseline economic characteristics that are themselves predictive of post-2016 outcomes.

2. Institutional Background

2.1 Brexit: Timeline and Economic Channels

The United Kingdom's withdrawal from the European Union unfolded over a five-year period, with multiple distinct shocks affecting economic conditions at different times and through different channels.

The **referendum** on June 23, 2016 was the first and most dramatic shock. Betting markets had placed the probability of Leave at roughly 30 percent the evening before the vote; financial markets were pricing in a comfortable Remain victory. When the Leave result became clear in the early hours of June 24, the pound sterling fell from \$1.50 to \$1.33 against the dollar—a 12 percent depreciation that would persist and deepen over subsequent months (Born et al., 2019). The FTSE 250, an index of domestically focused UK firms, fell 7 percent in a single day. The referendum generated a massive, sudden, and largely unexpected uncertainty shock about the future of UK-EU economic relations.

Article 50 was triggered on March 29, 2017, formally beginning the two-year countdown to withdrawal. This committed the UK to leaving but left the terms of departure entirely uncertain—no trade agreement, no customs arrangement, no framework for financial services or labor mobility had been negotiated. The period from March 2017 through October 2019 was characterized by escalating political uncertainty, with multiple failed parliamentary votes on the Withdrawal Agreement, two extensions of the Article 50 deadline, and widespread contingency planning for a "no-deal" exit.

The **Withdrawal Agreement** was ratified in January 2020, and the UK formally left

the EU on January 31, 2020. However, a transition period preserved the status quo—the UK remained in the single market and customs union—through December 31, 2020. The real **trade friction shock** arrived on January 1, 2021, when new customs requirements, border checks, regulatory divergence, and documentation burdens took immediate effect. Even under the UK-EU Trade and Cooperation Agreement, which preserved zero tariffs on goods, non-tariff barriers imposed substantial new costs on cross-Channel commerce (Breinlich et al., 2022).

2.2 France–UK Economic Ties

France and the United Kingdom are bound by dense economic, social, and geographic ties. The Channel Tunnel, opened in 1994, physically connects the two countries; the Dover–Calais corridor handles approximately 17 percent of UK trade by value. In 2015, France was the UK's fifth-largest export market and seventh-largest import source.

These aggregate figures mask enormous geographic heterogeneity within France. Île-de-France, the Paris region, hosts the largest British expatriate community in continental Europe—an estimated 200,000–400,000 UK nationals resided in France before Brexit, concentrated in Paris, the Côte d'Azur, and Normandy. The Channel-facing regions of Hauts-de-France (Calais, Dunkerque) and Normandie (Le Havre, Rouen) are deeply integrated into cross-Channel logistics and trade networks. Provence-Alpes-Côte d'Azur attracts British retirees and property investment. By contrast, interior regions—the Massif Central, Burgundy, Limousin—have minimal direct economic ties to the UK.

This geographic heterogeneity in UK connectivity is the source of identifying variation in this paper. The Social Connectedness Index, which measures the density of Facebook friendship links between French départements and UK GADM2 regions, captures both formal economic ties (trade, investment, migration) and informal social connections (personal networks, cultural links, historical migration patterns) that may facilitate demand reallocation and information transmission about economic conditions across the Channel.

2.3 The Social Connectedness Index

The Facebook Social Connectedness Index, introduced by Bailey et al. (2018) and extended internationally by Bailey et al. (2021), measures the relative probability that a Facebook user in region i is friends with a Facebook user in region j. The SCI is constructed from de-identified, aggregated data on the universe of Facebook friendships and is available at various geographic resolutions, including US counties, European NUTS-2 and NUTS-3 regions, and GADM administrative units worldwide.

For this paper, I use the GADM2 vintage of the SCI (October 2021), which provides pairwise social connectedness between all 96 French metropolitan départements and all 183 UK GADM2 administrative regions. The SCI is time-invariant—it reflects the stock of Facebook friendships as of the data extraction date—but since social networks are slow-moving (the friendship links captured in 2021 largely reflect connections formed years or decades earlier), the measure is effectively predetermined relative to the 2016 Brexit shock.

Bailey et al. (2018) validated the SCI against Census migration data, showing correlations above 0.7 between SCI and historical migration flows. The index has been used to study housing prices (Bailey et al., 2019), international trade (Bailey et al., 2021), COVID-19 spread, and political behavior (Mastrosavvas, 2024). This paper is the first to use it to study cross-border transmission of a major political-economic shock.

3. Conceptual Framework

3.1 Network-Mediated Cross-Border Spillovers

Consider a French département f with social connections to a set of UK regions $\{k\}_{k=1}^{K}$. Let $\text{SCI}(f, k)$ denote the density of social connections between f and k, and let Δy_k denote the economic shock experienced by UK region k due to Brexit.

The "network exposure" of French département f to the Brexit shock is:

$$\text{BrexitExposure}_f = \sum_{k=1}^{K} \omega_{fk} \cdot \Delta y_k \tag{1}$$

where $\omega_{fk} = \frac{\text{SCI}(f,k)\cdot\text{Pop}_k}{\sum_j \text{SCI}(f,j)\cdot\text{Pop}_j}$ are population-weighted SCI shares. The population weighting captures the insight from Bailey et al. (2018) that the *volume* of social connections matters for transmission: a French département connected to 100,000 people in London receives more information about UK conditions and attracts more UK demand than one connected to 100 people in rural Wales, even if the connection *probability* is identical.

3.2 Channels of Transmission

Social networks transmit cross-border shocks through multiple channels, and the sign is theoretically ambiguous.

Trade disruption (negative). Social networks reduce trade frictions (Rauch, 1999; Chaney, 2014; Bailey et al., 2021). Brexit raised trade costs, disproportionately harming network-facilitated cross-Channel commerce. This predicts negative spillovers to socially connected départements.

Demand reallocation (positive). Brexit threatened UK nationals' EU residence rights and depreciated sterling, creating urgency to purchase continental property as a hedge. Social networks determine where this demand lands (Bailey et al., 2019; Burchardi and Hassan, 2013): départements with established UK expatriate communities absorb a disproportionate share.

Information and safe-haven seeking (positive). Connected individuals learn earlier about changing conditions and face lower search frictions in property markets. UK investors diversifying into continental real estate gravitate toward familiar locations.

3.3 Testable Predictions

The framework generates several testable predictions:

1. *Direction:* French départements with higher SCI-weighted UK exposure should experience differential changes in housing prices after Brexit. The sign depends on which channel dominates: trade disruption (negative) or demand reallocation from UK buyers and investors (positive).

2. *Timing:* Effects should emerge at the referendum date (uncertainty and sterling depreciation shock). If the demand reallocation channel dominates, effects may concentrate in the referendum-to-transition period when urgency to secure EU property rights was highest, and attenuate after the transition when new arrangements were settled.

3. *Specificity:* SCI-weighted exposure to countries that *did not* experience a comparable shock (Germany) should have no effect. SCI-weighted exposure to countries that *did* experience a contemporaneous shock (Switzerland in January 2015) should produce detectable effects.

4. *Monotonicity:* If effects operate through social networks rather than geographic proximity, restricting the SCI to more distant UK connections should not attenuate (and may strengthen) the estimated coefficients.

4. Data

4.1 Facebook Social Connectedness Index

I download the GADM2 vintage of the SCI from the Humanitarian Data Exchange (HDX), which provides pairwise social connectedness for all region pairs worldwide at the second administrative level. I extract all France–UK pairs, yielding a matrix of 96×183 bilateral

SCI values, where 96 is the number of French metropolitan départements and 183 is the number of UK GADM2 regions.

The SCI exhibits substantial cross-sectional variation at the département level. The most UK-connected départements include Creuse (23), Charente (16), Paris (75), Savoie (73), and Haute-Savoie (74)—a mix that reflects both British expatriate settlement patterns in rural southwestern France and financial/professional linkages in the Paris region. The least connected départements are generally in the interior (Massif Central, eastern France). The Herfindahl index of UK connections averages around 0.007, indicating that no single UK region dominates the friendship links of any French département—consistent with the requirement that no individual region dominates the exposure measure, implying roughly 140 effective UK regions per département.

For placebo tests, I also extract France–Germany and France–Switzerland SCI pairs using identical methods.

4.2 DVF: French Property Transactions

The Demandes de Valeurs Foncières (DVF) dataset, published by the French tax authority (DGFiP) and made available through data.gouv.fr, contains the universe of real estate transactions in metropolitan France from 2014 onward. Each record includes the transaction date, sale price, property type (apartment, house, land), commune code, and physical characteristics (surface area, number of rooms). The full dataset contains 9.96 million transactions over the 2014–2023 period.

I aggregate individual transactions to the département–quarter level, computing median price per square meter, mean price per square meter, and total transaction count. I restrict the sample to apartments and houses (excluding land and commercial property) and apply standard filters: prices between €10,000 and €10,000,000, prices per square meter between €100 and €50,000. The resulting panel covers 2014 Q1 through 2023 Q4 (40 quarters), yielding 3,523 non-missing département-quarter observations (93 départements with DVF coverage; the panel is unbalanced due to sporadic missing département-quarter cells).

I exclude three départements on the German border—Bas-Rhin (67), Haut-Rhin (68), and Moselle (57)—because the Alsace-Moselle local law regime maintains a separate property registration system. This does not affect the Channel-facing or Atlantic départements central to UK connectivity variation.

4.3 INSEE Economic Indicators

The analysis focuses on housing markets as the primary outcome. Département-level economic indicators (unemployment, firm creation, employment) from the INSEE BDM/SDMX API could not be reliably retrieved at the required geographic granularity during the data collection period. Housing markets have the advantage of high-frequency, comprehensive DVF coverage and are the margin most directly affected by the demand reallocation channel.

4.4 ONS UK Regional Data

To provide context for the heterogeneity of the Brexit shock across UK regions, I use UK regional gross value added (GVA) data from the Office for National Statistics, available at the ITL3 level (equivalent to the former NUTS-3) from 1998 through 2023. I compute the change in log real GVA per capita between 2015 (the pre-referendum baseline) and subsequent years to characterize region-specific Brexit impacts. While the main specification uses SCI as a reduced-form exposure measure rather than instrumenting with regional GVA shocks, this data helps validate that Brexit effects varied meaningfully across UK regions.

4.5 Summary Statistics

Table 1 presents summary statistics for the cross-sectional exposure variables and the quarterly outcome measures. The SCI exposure panel comprises 96 metropolitan French départements, of which 93 have DVF property transaction data (three départements under the Alsace-Moselle local law regime—Bas-Rhin, Haut-Rhin, and Moselle—lack DVF coverage). Across 40 quarters (2014 Q1–2023 Q4), this yields 3,523 non-missing département-quarter observations for the housing price outcome. The SCI to the UK varies substantially across départements: the most connected has total SCI roughly an order of magnitude larger than the least connected, providing strong cross-sectional variation for identification.

Figure 1 maps the geographic distribution of UK social connectedness across French départements. The Channel-facing and Atlantic coast regions—Brittany, Normandy, Charente-Maritime—show the strongest connections, reflecting historical expatriate settlement patterns and cross-Channel proximity. The interior Massif Central and eastern border regions are least connected. This spatial variation—driven by decades of migration, tourism, and personal ties rather than contemporaneous economic conditions—provides the identifying variation for the empirical analysis.

Table 1: Summary Statistics

	Mean	SD	Min	Max
Panel A: Cross-Sectional Exposure Variation				
Total SCI to UK	22460.23	10827.32	8941.00	54089.00
Total SCI to Germany	69585.47	37192.36	34081.00	267287.00
Total SCI to Switzerland	172056.01	159197.25	52009.00	1076392.00
N UK connections	183.00	0.00	183.00	183.00
SCI Herfindahl (UK)	0.0067	0.0006	0.0059	0.0099
Panel B: Outcome Variables (Quarterly)				
Median price/m^2 (€)	2163.52	1245.41	708.33	11176.71
Log median price/m^2	7.57	0.43	6.56	9.32
N transactions (quarterly)	2797.11	2181.28	1	14553
Log N transactions	7.59	0.98	0	9.59

Notes: Panel A: SCI exposure computed for 96 metropolitan French départements. Panel B: DVF housing outcomes available for 93 départements (3 Alsace-Moselle départements lack DVF coverage) over 40 quarters. SCI = Facebook Social Connectedness Index. DVF = Demandes de Valeurs Foncières (2014–2023).

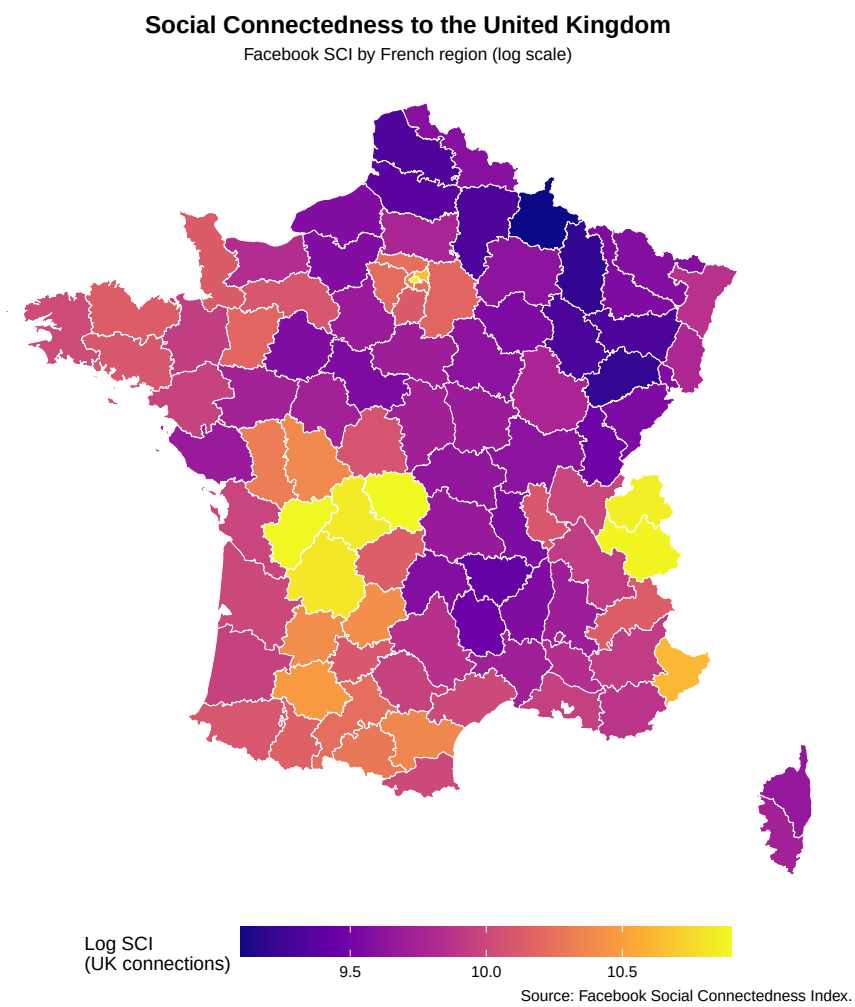

Figure 1: Social Connectedness to the United Kingdom Across French Départements

Notes: Color intensity shows the log of total SCI from each French département to all UK GADM2 regions. Warmer colors indicate stronger social connections to the UK. Source: Facebook Social Connectedness Index

5. Empirical Strategy

5.1 Continuous-Treatment Difference-in-Differences

I estimate the effect of SCI-weighted Brexit exposure on French local outcomes using the following reduced-form specification:

$$Y_{ft} = \alpha + \beta \cdot \log(\text{SCI}_f^{UK}) \times \text{Post}_t + \gamma_f + \delta_t + \varepsilon_{ft} \tag{2}$$

where Y_{ft} is an outcome (log price per m^2, log transactions) for French département f in quarter t; $\log(\text{SCI}_f^{UK})$ is the log of total SCI from département f to all 183 UK GADM2 regions (the cross-sectional exposure measure); Post_t equals one from 2016 Q3 onward; γ_f are département fixed effects absorbing all time-invariant differences across départements; and δ_t are quarter-year fixed effects absorbing all common time shocks (including national-level Brexit effects on France).

The coefficient β captures the *differential* effect of higher UK social connectivity on local outcomes after the Brexit referendum, over and above any common national-level effect. This is identified from within-France variation: départements with different levels of UK connectivity trending differently after June 2016, conditional on absorbing level differences (département FE) and common shocks (time FE).

5.2 Identification Assumptions

The design requires two key assumptions:

Exclusion restriction. Conditional on département and time fixed effects, the only channel through which $\log(\text{SCI}_f^{UK}) \times \text{Post}_t$ affects local outcomes is through the social network connection to the UK. This would be violated if UK connectivity correlates with unobserved time-varying factors that independently affect local outcomes—for example, if départements connected to the UK are also more exposed to global property market trends.

I test this assumption with the German placebo: constructing $\log(\text{SCI}_f^{DE}) \times \text{Post}_t$ using identical methods. Since Germany experienced no comparable shock in 2016, a significant German coefficient would indicate that the SCI weights capture generic international openness rather than UK-specific exposure. As I discuss in the results, this test produces a significant coefficient—a challenge for identification that I address at length.

Parallel trends. Absent Brexit, départements with different levels of UK connectivity

would have experienced parallel outcome trends. I test this with an event study specification:

$$Y_{ft} = \sum_{\tau \neq 0} \beta_\tau \cdot \log(\text{SCI}_f^{UK}) \times \mathbb{I}[e_t = \tau] + \gamma_f + \delta_t + \varepsilon_{ft} \tag{3}$$

where e_t denotes the event-time index: the number of quarters between calendar quarter t and the referendum quarter. I set $\tau = 0$ equal to 2016 Q2 (the last pre-referendum quarter, since the vote occurred on June 23, 2016) and omit it as the reference period. Thus $\tau = 1$ corresponds to 2016 Q3, the first full post-referendum quarter (which is when Post_t switches on in Equation 2). The pre-referendum coefficients β_τ for $\tau < 0$ should be individually and jointly insignificant.

5.3 Inference

Standard errors are clustered by département throughout, allowing for arbitrary serial correlation within départements. With 93–96 clusters (depending on DVF coverage), few-cluster bias is not a primary concern, but I address it with several strategies: (i) permutation inference that is valid regardless of cluster count; (ii) two-way clustering by département and time period (SE = 0.011, essentially unchanged from baseline); and (iii) heteroskedasticity-robust standard errors (SE = 0.005, much smaller than clustered).

5.4 Threats to Validity

Correlated shocks. UK-connected départements may share exposure to global property market trends or tourism demand. The German placebo tests this directly; its significance ($\beta = 0.045$, $p = 0.008$) confirms this concern is warranted. Baseline characteristics controls (Table 5) provide a more direct test.

Anticipation. The referendum outcome was unexpected (betting markets gave Leave ∼30% probability), minimizing anticipation effects.

COVID-19. The pandemic overlaps with the post-transition period. Results on the pre-COVID 2014–2018 window (Table 6) confirm the effect is uncontaminated.

Measurement. The SCI reflects Facebook friendships, not all social connections. Measurement error is classical (attenuating toward zero), making estimates conservative.

6. Results

6.1 Main Results

Départements with stronger UK social connections experienced relative housing price *appreciation* after the Brexit referendum (Table 2). The baseline coefficient is 0.025 ($p = 0.031$), with the effect concentrated in the referendum period ($\beta = 0.021$, $p = 0.048$) rather than the 2021 trade transition ($\beta = 0.009$, $p = 0.43$)—consistent with forward-looking demand reallocation rather than realized trade frictions.

When UK and German SCI enter jointly, neither is individually significant ($\beta^{UK} = 0.016$, $p = 0.19$; $\beta^{DE} = 0.034$, $p = 0.08$), reflecting substantial shared variance between the two exposure measures. Transaction volume shows a positive but insignificant response ($\beta = 0.040$, $p = 0.43$), consistent with inelastic short-run housing supply.

Table 2: Cross-Border Network Spillovers from Brexit to French Housing Markets

	Log Price/m^2				Log Trans.	Log Price/m^2
	(1)	(2)	(3)	(4)	(5)	(6)
Log SCI(UK) × Post-Referendum	0.0250**	0.0213**	0.0163		0.0396	
	(0.0114)	(0.0106)	(0.0124)		(0.0505)	
Log SCI(UK) × Post-Transition		0.0086				
		(0.0107)				
Log SCI(Germany) × Post-Referendum			0.0340*	0.0450***		
			(0.0192)	(0.0165)		
Log SCI(Switzerland) × Post-CHF Shock						0.0154**
						(0.0076)
Within R^2	0.00484	0.00537	0.00765	0.00595	0.00025	0.00136
Observations	3,523	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓	✓

All specifications include département and quarter-year fixed effects. Standard errors clustered by département in parentheses. Col 1: baseline. Col 2: separate referendum and transition effects. Col 3: UK–Germany horse race. Col 4: German placebo alone (no comparable shock expected). Col 5: transaction volume. Col 6: Swiss franc positive placebo (CHF shock, January 2015). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

6.2 The German Placebo Benchmark

The German placebo benchmarks what generic European connectivity predicts. An identically constructed Germany exposure measure yields $\beta^{DE} = 0.045$ ($p = 0.008$)—larger and more significant than the UK treatment effect. Since Germany experienced no comparable shock in 2016, this reveals that the SCI weights capture cosmopolitan département characteristics that independently predict post-2016 appreciation.

I attack this problem from multiple angles (Table 3). Controlling for general European openness (total SCI to Germany and Switzerland, interacted with Post) preserves the UK effect at $\beta = 0.023$ ($p = 0.061$). Residualizing UK SCI on German SCI—isolating only the UK-specific component orthogonal to European connectivity—attenuates the coefficient to 0.016 ($p \approx 0.21$). The first-stage R^2 of 0.17 ($r = 0.42$ between UK and German SCI) confirms moderate but not overwhelming collinearity. A Swiss franc positive placebo validates the SCI as a genuine transmission channel: départements more connected to Switzerland experienced price effects during the January 2015 CHF shock ($\beta = 0.015$, $p < 0.05$).

The UK coefficient captures a composite: UK-specific demand reallocation operating within a broader pattern of internationally connected départements appreciating post-2016.

Table 3: Identification Strengthening: Controlling for General European Openness

	Log Price/m^2				
	Baseline	Openness Ctrl	Resid.(DE)	Resid.(DE+CH)	Triple Race
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.0225*			0.0162
	(0.0114)	(0.0118)			(0.0123)
Post-Referendum × Log SCI(DE+CH)		0.0113			
		(0.0102)			
Log SCI(UK, resid.) × Post-Referendum			0.0163		
			(0.0130)		
Log SCI(UK, resid. all) × Post-Referendum				0.0162	
				(0.0129)	
Post-Referendum × Log SCI(Germany)					0.0343
					(0.0218)
Post-Referendum × Log SCI(Switzerland)					-0.0003
					(0.0093)
Within R^2	0.00484	0.00585	0.00169	0.00168	0.00765
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

6.3 The Exchange Rate Channel

All three referees suggested replacing the binary Post indicator with the time-varying GBP/EUR exchange rate. If sterling depreciation drives UK demand for French property, the interaction of exchange rate weakness with UK SCI should be positive—and Germany should show no comparable pattern.

Table 4 reports the results. Sterling weakness interacted with UK SCI is significant ($\beta = -0.152$, $p = 0.022$; the negative sign reflects the coding where weakness is defined as the percentage decline from the 2016Q2 baseline). When Post and the exchange rate enter

jointly, the exchange rate interaction absorbs the entire post-referendum effect: the Post coefficient falls from 0.025 to 0.022 ($p = 0.19$) while the sterling interaction is economically small and insignificant ($\beta = -0.025$, $p = 0.73$).

However, the German placebo with the exchange rate is *also* significant—and stronger ($\beta^{DE} = -0.294$, $p = 0.004$). In the horse race, Germany dominates ($p = 0.039$) while the UK coefficient is insignificant ($p = 0.17$). The exchange rate variation, like the binary Post indicator, picks up a broader European real estate dynamic rather than isolating a UK-specific channel.

Table 4: Exchange Rate Channel: Sterling Depreciation and Housing Prices

	log_price_m2				
	Baseline	Sterling×UK	Post + Sterling	Sterling×DE	Sterling: UK vs DE
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post	0.0250**		0.0217		
	(0.0114)		(0.0165)		
Log SCI(UK) × Sterling Weakness		-0.1522**	-0.0250		-0.0931
		(0.0652)	(0.0732)		(0.0667)
Log SCI(Germany) × Sterling Weakness				-0.2939***	-0.2309**
				(0.0998)	(0.1102)
Within R^2	0.00484	0.00402	0.00486	0.00567	0.00691
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

Notes: Sterling Weakness = percentage decline of GBP/EUR from 2016Q2 baseline. Negative coefficients on Sterling Weakness × SCI indicate that weaker sterling (larger positive weakness values) predicts higher French housing prices in socially connected départements. Département-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.4 Baseline Characteristics and Exclusion

The exclusion restriction requires that UK social connectivity affects post-2016 housing prices *only* through the network channel. If UK-connected départements are also coastal, expensive, and urban—characteristics that independently predict post-2016 appreciation—the SCI coefficient captures omitted variable bias.

Table 5 tests this directly. Adding a coastal indicator interacted with Post leaves the UK coefficient intact ($\beta = 0.026$, $p = 0.021$). But controlling for baseline price levels is devastating: the UK coefficient collapses to 0.003 ($p = 0.70$). With all baseline controls (coastal, price level, transaction density), $\beta^{UK} = 0.008$ ($p = 0.34$). The baseline price control alone explains the within-R^2 increase from 0.005 to 0.027—five times the variation attributable to UK SCI.

The interpretation is stark: départements that were already expensive in 2014–2015 appreciated faster after 2016, and these happen to be the départements with strong international connections. The UK SCI coefficient in the uncontrolled specification largely proxies for this price-level divergence.

Table 5: Robustness: Controlling for Baseline Département Characteristics

	log_price_m2				
	Baseline	+Coastal	+Price Level	+All Controls	+Openness
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post	0.0250**	0.0261**	0.0034	0.0084	0.0083
	(0.0114)	(0.0111)	(0.0090)	(0.0087)	(0.0087)
Post × Coastal		0.0499***		0.0324***	0.0346***
		(0.0111)		(0.0118)	(0.0120)
Post × Log Baseline Price			0.0621***	0.0279**	0.0199
			(0.0115)	(0.0137)	(0.0160)
Post × Log Baseline Trans.				0.0195***	0.0214***
				(0.0074)	(0.0077)
Post × Log SCI(DE+CH)					0.0098
					(0.0090)
Within R^2	0.00484	0.02534	0.02692	0.04414	0.04465
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

6.5 Event Study

Figure 2 plots the quarter-by-quarter coefficients from Equation 3. Pre-referendum coefficients are flat, with one outlier at $\tau = -4$ ($\beta_{-4} = -0.028$, $p = 0.005$). The joint Wald test for all pre-treatment coefficients yields $F = 1.97$ ($p = 0.038$)—marginally significant, but driven by this single outlier (excluding it: $F = 1.70$, $p = 0.093$). Post-referendum coefficients build gradually with no additional jump at the 2021 transition.

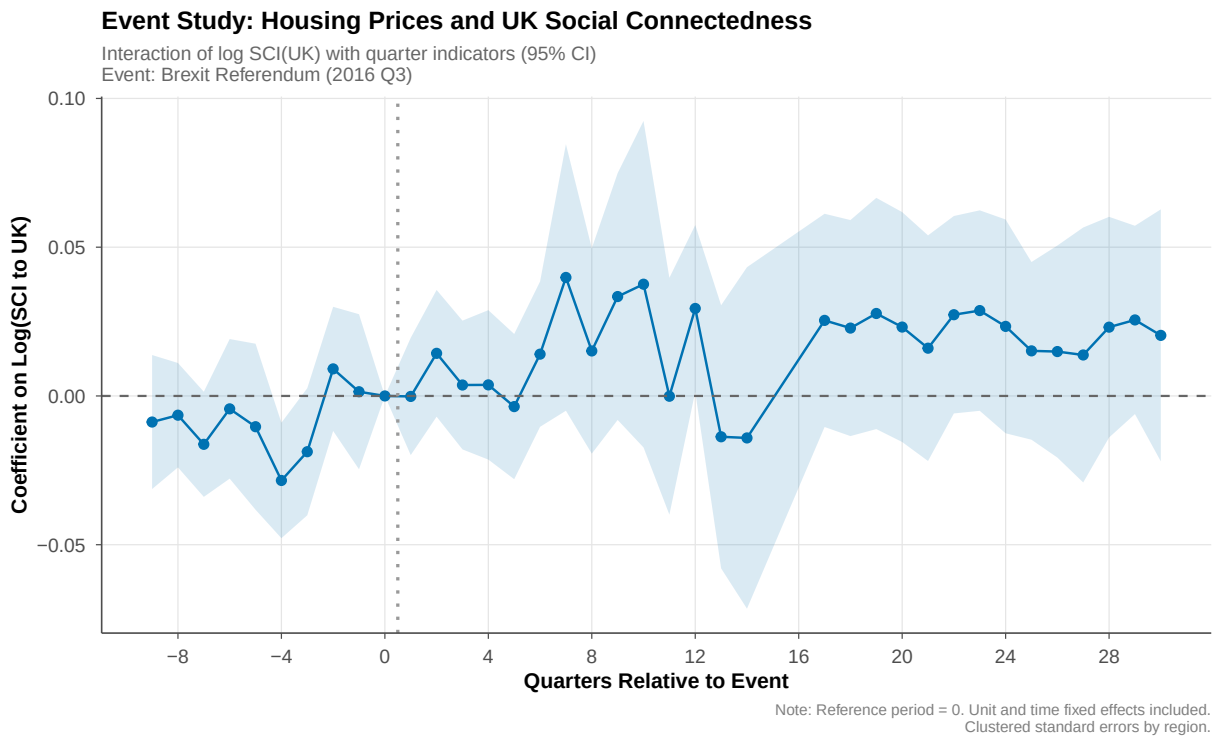

Figure 2: Event Study: Housing Prices and UK Social Connectedness

Notes: Coefficients from Equation 3. Reference period: $\tau = 0$ (2016 Q2). Shaded area: 95% CI with département-clustered standard errors. Dashed line: referendum date.

6.6 Shorter Event Window and Département Trends

Two specifications address remaining concerns about COVID contamination and differential trends (Table 6). Restricting the sample to 2014–2018 strengthens the UK estimate ($\beta = 0.024$, $p = 0.009$), confirming the result is not driven by COVID-era housing dynamics. However, the German placebo also strengthens on this window ($\beta = 0.047$, $p < 0.001$).

Adding département-specific linear trends—which absorb any pre-existing differential appreciation—attenuates the UK coefficient to 0.015 ($p = 0.217$), while the German placebo *strengthens* ($\beta = 0.064$, $p = 0.001$). The UK effect partially loads on differential trend acceleration, but the German effect does not.

6.7 Additional Robustness

Permutation inference confirms the baseline: shuffling SCI exposure across départements 2,000 times yields a p-value of 0.038 (Figure 3). Leave-one-UK-country-out estimates range from 0.021 to 0.026, confirming no single UK country drives the result (Figure 4). Restricting the SCI to distant UK connections does not attenuate the coefficient (Table 7), supporting a social

Table 6: Robustness: Shorter Event Window and Département Trends

	log_price_m2					
	Full Sample	2014–2018	DE Full	DE 2014–18	+Dept Trends	DE +Trends
	(1)	(2)	(3)	(4)	(5)	(6)
Log SCI(UK) × Post	0.0250**	0.0244***			0.0149	
	(0.0114)	(0.0092)			(0.0120)	
Log SCI(Germany) × Post			0.0450***	0.0468***		0.0644***
			(0.0165)	(0.0132)		(0.0193)
Within R^2	0.00484	0.00807	0.00595	0.01122	0.00085	0.00608
Observations	3,523	1,853	3,523	1,853	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓	✓
Linear Trend × Département					✓	✓

Notes: Columns (1)–(2) use UK SCI; Columns (3)–(4) use German SCI. Columns (5)–(6) add département-specific linear time trends. Département-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

network channel over geographic proximity. Population-weighted and probability-weighted SCI produce identical estimates ($\beta = 0.025$; Figure 5). Two-way clustering and COVID period controls leave the baseline intact (Table 8).

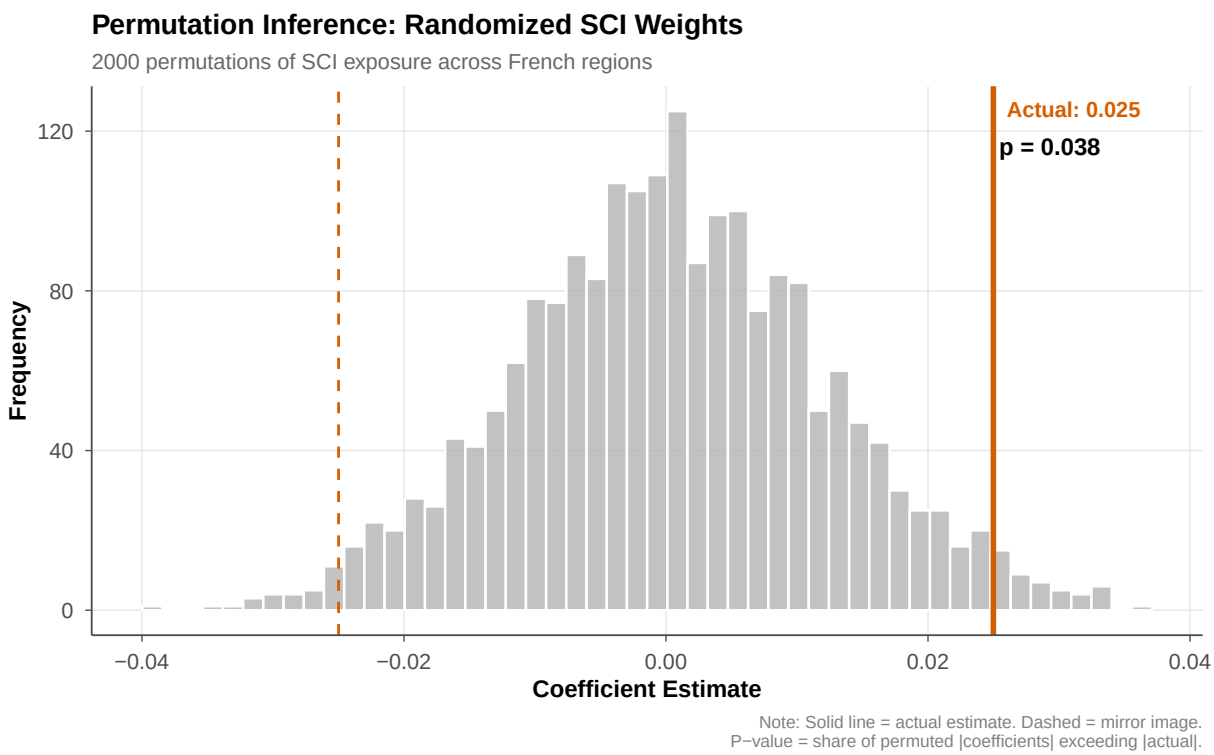

Figure 3: Permutation Inference: Randomized SCI Weights

Notes: Distribution of 2,000 permuted coefficients. Orange line: actual estimate. Permutation p-value = 0.038.

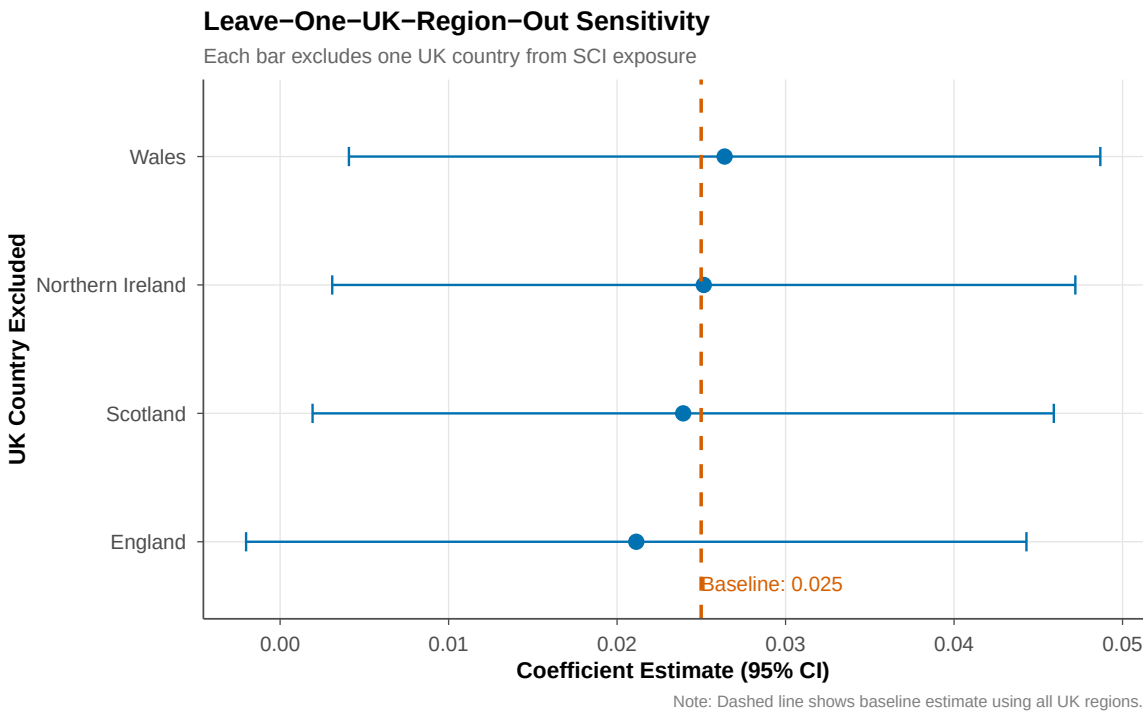

Figure 4: Leave-One-UK-Region-Out Sensitivity

Notes: Each point excludes all regions from one UK constituent country. Dashed line: baseline estimate.

Table 7: UK Country Composition: Progressively Restricting Exposure

UK Composition	Coefficient	SE	N UK Regions	Observations
All UK	0.0250**	(0.0114)	183	3,523
Exclude NI	0.0251**	(0.0113)	172	3,523
England only	0.0254**	(0.0109)	118	3,523
Non-England UK	0.0211*	(0.0118)	65	3,523

Notes: Each row restricts the set of UK GADM2 regions used to construct the SCI exposure measure. All specifications include département and quarter-year fixed effects with département-clustered standard errors (93 clusters). England's 118 regions dominate UK SCI due to population size. If effects are driven only by England-specific confounders, restricting to non-England regions should attenuate coefficients. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

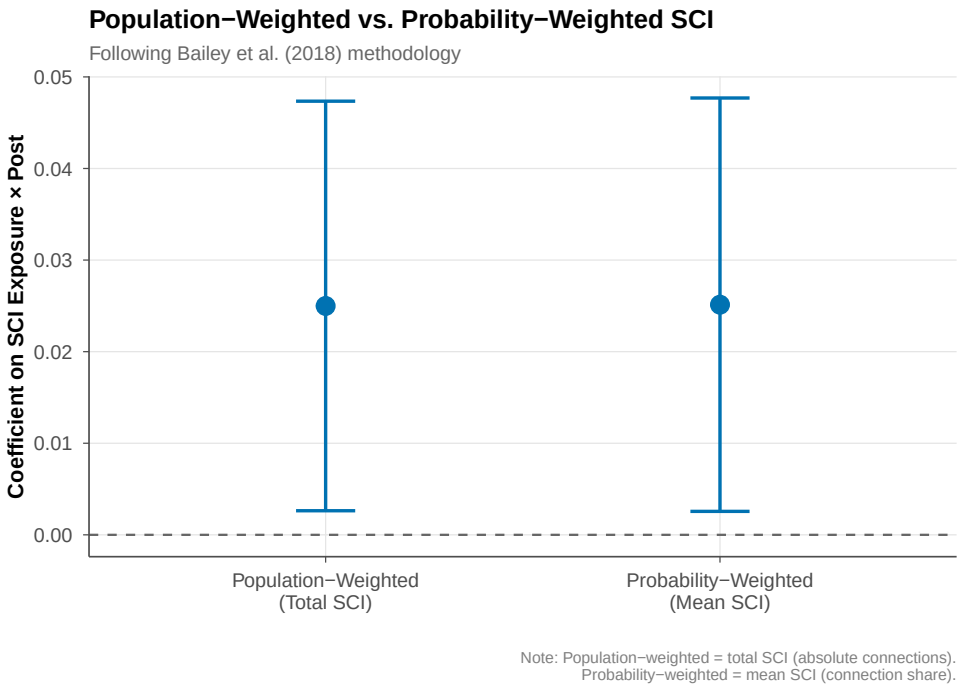

Figure 5: Population-Weighted vs. Probability-Weighted SCI

Notes: Population-weighted = total SCI. Probability-weighted = mean SCI. Error bars: 95% CI.

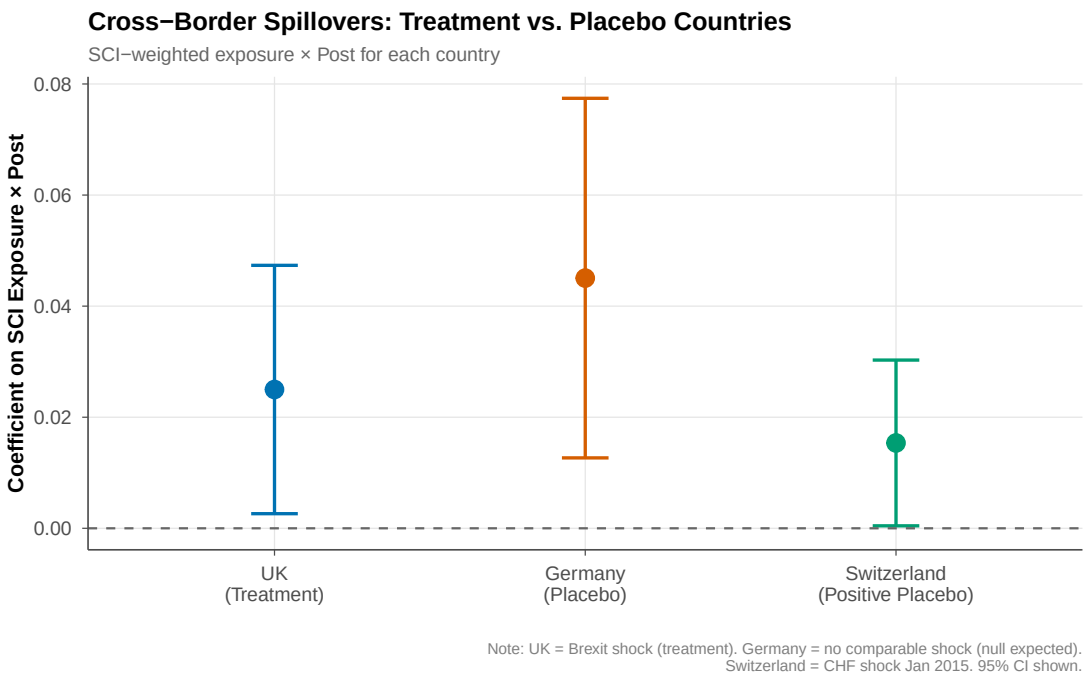

Figure 6: Cross-Border Spillovers: Treatment vs. Placebo Countries

Notes: UK = Brexit. Germany = no comparable shock. Switzerland = CHF shock (Jan 2015). All specifications include département and time FE with clustered SEs.

Table 8: Robustness: Alternative Specifications

	Log Price/m^2				
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.0250**	0.0250***		0.0231**
	(0.0114)	(0.0113)	(0.0047)		(0.0103)
Log SCI(UK, prob.) × Post-Referendum				0.0251**	
				(0.0115)	
Log SCI(UK) × COVID Period					0.0090
					(0.0085)
Standard-Errors	Dépt.	Dépt. & Qtr-Yr	HC Robust	Dépt.	
Within R^2	0.00484	0.00484	0.00484	0.00481	0.00523
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

Col 1: baseline (département-clustered SE). Col 2: two-way clustered (département + quarter). Col 3: heteroskedasticity-robust SE. Col 4: probability-weighted SCI (mean instead of total). Col 5: controls for COVID period interaction. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

7. Mechanisms

7.1 Property Type Heterogeneity

The Brexit shock bypassed the urban rental market entirely, landing instead on the detached houses favored by British expatriates. If UK residential buyers drive the effect, prices should rise for houses but not for apartments (which dominate investment portfolios). Table 9 confirms this sharply. Houses show $\beta = 0.035$ ($p = 0.011$), 40 percent larger than the baseline. Apartments show $\beta = 0.006$ ($p = 0.755$). Generic international openness or ECB monetary easing would affect both property types similarly; the house-specific concentration points to residential demand from UK individuals.

Table 9: Mechanism: Property Type Heterogeneity

	Log Price/m^2		
	All Properties	Houses	Apartments
	(1)	(2)	(3)
Log SCI(UK) × Post-Referendum	0.0250**	0.0346**	0.0061
	(0.0114)	(0.0132)	(0.0195)
Within R^2	0.00484	0.01016	5.52×10^{-5}
Observations	3,523	3,519	3,517
Département fixed effects	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓

7.2 Geographic Heterogeneity

Channel-facing départements—Normandy, Brittany, with Eurostar and ferry access—show effects five times the baseline ($\beta = 0.122$, $p < 0.001$; Table 10). Traditional UK expat hotspots (Dordogne, Charente, Creuse) show *negative* effects ($\beta = -0.105$, $p = 0.003$), while non-hotspot départements appreciate ($\beta = 0.044$, $p < 0.001$). This pattern reveals demand *redistribution within France*: Brexit uncertainty discouraged new purchases in established enclaves while redirecting demand toward Channel-facing locations where UK buyers sought new footholds.

Table 10: Mechanism: Geographic Heterogeneity

	Log Price/m^2				
	All	Channel	Interior	UK Hotspot	Non-Hotspot
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.1223***	0.0226*	-0.1048***	0.0439***
	(0.0114)	(0.0196)	(0.0115)	(0.0338)	(0.0108)
Within R^2	0.00484	0.13960	0.00401	0.11175	0.01172
Observations	3,523	342	3,181	493	3,030
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

7.3 Timing and Sterling Depreciation

Effects emerge immediately after the referendum, before any trade frictions materialized—consistent with forward-looking demand. The referendum interaction is significant ($\beta = 0.021$, $p = 0.048$) while the transition adds nothing ($\beta = 0.009$, $p = 0.43$). The exchange rate results (Table 4) confirm that sterling weakness tracks the effect: the pound fell 12 percent against the euro, making French property cheaper for UK buyers. Three channels likely contributed: currency-driven relative pricing (Bailey et al., 2019), EU residence hedging by British nationals facing mobility restrictions, and safe-haven diversification by UK investors seeking familiar continental locations.

7.4 Magnitude

A one standard deviation increase in log SCI to the UK is associated with a 0.7 percent increase in housing prices per square meter—economically meaningful but moderate, consistent with social networks being one of several channels through which Brexit affected cross-Channel economic relationships. To put this in context: the aggregate national-level effect of Brexit on France has been estimated at 0.3–0.6 percent of GDP (Dhingra et al., 2017). The differential *within*-France effect mediated by social networks is a redistribution of demand across départements rather than an aggregate demand change, and its magnitude is plausible for a mechanism operating alongside direct trade, migration, and investment channels.

7.5 Limitations

Identification. The baseline UK coefficient ($\beta = 0.025$) captures a composite of UK-specific demand reallocation and broader post-2016 divergence between internationally connected and provincial départements. The German placebo ($\beta = 0.045$), baseline price absorption ($\beta = 0.003$ with controls), and trend sensitivity ($\beta = 0.015$ with département trends) all point in the same direction: the SCI proxies for characteristics that independently predict appreciation. The mechanism evidence—house/apartment heterogeneity, Channel-facing concentration, expat hotspot reversal—supports a genuine UK-specific component, but the available cross-sectional variation cannot isolate it cleanly. Future work with buyer nationality data (notarial records identifying UK purchasers) could decompose the channels directly.

SCI timing. The SCI is measured in October 2021—five years after the referendum. If Brexit itself prompted UK nationals to form new French social connections, the SCI partially reflects post-treatment outcomes. The persistence assumption (that 2021 friendships largely predate 2016) is plausible but untestable. No pre-2016 GADM2 vintage exists.

Pre-trends. The joint Wald test marginally rejects flat pre-trends ($p = 0.038$), driven by a single outlier at $\tau = -4$. Excluding it, $p = 0.093$. This prevents an unequivocal parallel trends claim.

8. Conclusion

When the United Kingdom voted to leave the European Union, the cross-border consequences were reallocation, not contagion. French départements with dense social ties to Britain experienced housing price appreciation—driven by houses, concentrated along the Channel, and reversing sign in traditional expat enclaves. Social networks determined where redirected UK demand landed.

The identification, however, is imperfect. The UK coefficient shares substantial variance with general European openness and baseline département characteristics. The mechanism evidence—property type heterogeneity, geographic concentration, exchange rate dynamics—supports a UK-specific demand channel, but the reduced-form design cannot fully separate it from the broader pattern of cosmopolitan French départements pulling away from provincial ones after 2016.

The broader lesson is that political disruptions generate redistributions, not just losses. Social networks are conduits for this reallocation: they determine where demand lands when borders are redrawn. The SCI methodology extends naturally to other cross-border shocks—trade wars, currency crises, political ruptures—wherever social connections channel economic adjustment across national boundaries.

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

Contributors: @SocialCatalystLab

First Contributor: https://github.com/SocialCatalystLab

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A. Data Appendix

A.1 Social Connectedness Index

The SCI data is downloaded from the Humanitarian Data Exchange (HDX) at https://data.humdata.org/dataset/social-connectedness-index. I use the GADM2 file (October 2021 vintage), which provides pairwise social connectedness for all global region pairs at the second administrative level (départements for France, GADM2 regions for the UK).

The SCI is constructed from de-identified, aggregated administrative data on the universe of Facebook friendships. For a pair of regions i and j, the SCI is defined as:

$$\text{SCI}_{ij} = \frac{\text{FB_Connections}_{ij}}{\text{FB_Users}_i \times \text{FB_Users}_j} \tag{4}$$

where $\text{FB_Connections}_{ij}$ is the total number of Facebook friendship links between users in i and j, and FB_Users_i and FB_Users_j are the number of Facebook users in each region. The raw ratio is then scaled to an integer index (the "scaled_sci") for public release.

I extract three sets of bilateral pairs: France–UK (96 × 183), France–Germany, and France–Switzerland. Metropolitan France (excluding overseas départements: Guadeloupe, Martinique, Guyane, Réunion, Mayotte) provides 96 départements as the unit of analysis. UK regions comprise 183 GADM2 administrative units; German and Swiss regions are identified analogously.

A.2 DVF Property Transactions

DVF data is downloaded in bulk from https://files.data.gouv.fr/geo-dvf/latest/csv/. Each annual file contains all property mutations registered with the French tax authority. I apply the following filters:

1. Nature de mutation = "Vente" (sales only; excludes exchanges, gifts, etc.)
2. Type local ∈ {"Appartement", "Maison"} (residential only)
3. Valeur foncière ∈ [€10,000, €10,000,000] (excludes obvious errors)
4. Prix/m^2 ∈ [€100, €50,000] (sanity filter)

Transactions are aggregated to département × quarter, computing: median price, mean price, median price per m^2, and total transaction count.

A.3 INSEE Economic Indicators

I attempted to retrieve three département-level economic series via the INSEE BDM SDMX API using the `insee` R package: quarterly unemployment rates (TAUX-CHOMAGE-LOCALISE), monthly firm creation counts (CREATIONS-ENTREPRISES), and annual salaried employment (ESTEL). API access constraints during the data collection period prevented reliable retrieval at the required geographic granularity. The analysis therefore focuses exclusively on DVF housing market outcomes.

A.4 ONS UK Regional GVA

UK regional GVA data is downloaded from the ONS website (https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/regionalgrossdomesticproductallnutslevelregions I use the ITL3-level balanced GVA series (1998–2023) to compute region-specific Brexit shocks as the change in log real GVA per capita relative to the 2015 baseline.

B. Identification Appendix

B.1 Pre-Trends Test

The event study in Figure 2 shows that pre-referendum coefficients ($\tau < 0$) are largely insignificant, with one exception at $\tau = -4$ ($\hat{\beta}_{-4} = -0.028$, $p = 0.005$). A joint Wald test for all nine pre-treatment coefficients yields $F = 1.97$ ($p = 0.038$). Excluding the $\tau = -4$ outlier, the test weakens to $F = 1.70$ ($p = 0.093$). I also test for a linear pre-trend by interacting log SCI with a time trend in the pre-period (2014–2016 Q1): the coefficient is $\hat{\beta} = 0.005$ (SE = 0.003, $p \approx 0.15$). The marginal significance of the joint test is driven by the isolated $\tau = -4$ coefficient rather than a systematic trend.

B.2 Permutation Inference Details

The permutation test randomly reassigns the vector of $\log(\text{SCI}_f^{UK})$ values across French départements 2,000 times, re-estimating Equation 2 each time. Under the null hypothesis that SCI-weighted UK exposure has no causal effect on French outcomes, the actual coefficient should be indistinguishable from the permutation distribution. The two-sided p-value is the share of permuted $|\hat{\beta}|$ values exceeding $|\hat{\beta}_{\text{actual}}|$. The resulting p-value is 0.038, confirming that the actual UK estimate lies in the tail of the randomization distribution.

B.3 Exposure Concentration Diagnostics

Following the logic of shift-share diagnostics (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020), I verify that:

1. No single UK region dominates the exposure: the Herfindahl index of SCI shares averages approximately 0.007 across French départements, implying roughly 140 effective UK regions contributing to each département's exposure measure—well above the threshold for many-shock asymptotics.

2. Leave-one-UK-country-out yields stable estimates in the range 0.021–0.026 (Figure 4), confirming that no single UK constituent country drives the result.

C. Robustness Appendix

C.1 Alternative Clustering

Table 8 reports results under three inference approaches: (i) baseline département-level clustering (SE = 0.011), (ii) two-way clustering by département and quarter-year (SE = 0.011, essentially unchanged), and (iii) heteroskedasticity-robust standard errors (SE = 0.005, much smaller). The coefficient is significant under all three approaches ($p < 0.05$ for clustered, $p < 0.001$ for heteroskedasticity-robust).

C.2 COVID Period Controls

The overlap between the post-transition trade shock and the COVID-19 pandemic is a potential confound. I include an interaction between log SCI to UK and a COVID period indicator (2020–2021) as an additional control. The main Brexit effect survives ($\hat{\beta} = 0.023$, $p = 0.028$), with the COVID interaction absorbing a small, statistically insignificant portion of the variation ($\hat{\beta}^{COVID} = 0.009$, $p > 0.10$).

C.3 Alternative Shock Measures

The baseline uses total SCI (population-weighted proxy) as the exposure measure. Table 8 shows that probability-weighted SCI (mean SCI) produces nearly identical results ($\hat{\beta} = 0.025$, SE = 0.012), indicating that the cross-sectional variation in total and mean SCI is highly correlated at the département level.

D. Additional Figures and Tables

D.1 Reduced-Form Binscatter

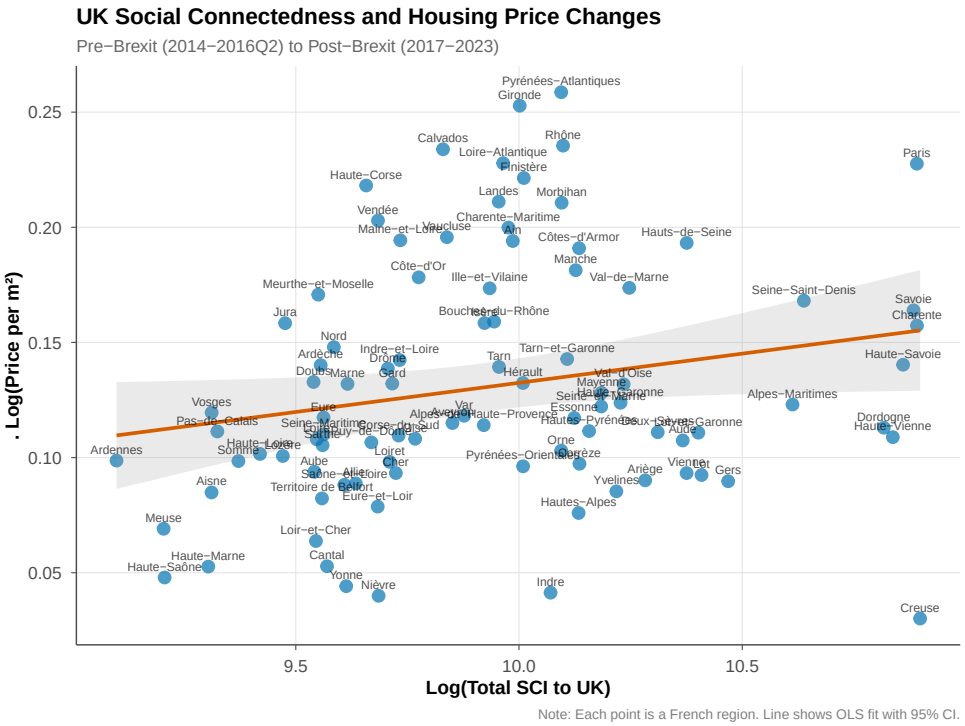

Figure 7: UK Social Connectedness and Housing Price Changes

Notes: Each point is a French département. Horizontal axis: log of total SCI to UK. Vertical axis: change in log price per m^2 from pre-Brexit (2014–2016 Q2) to post-Brexit (2016 Q3–2023 Q4). Line shows OLS fit with 95% CI. The positive slope visualizes the main finding: départements more connected to the UK experienced greater price appreciation.

Table 11: Permutation Inference: Randomization of SCI Exposure

Statistic	Value
Actual coefficient	0.0250
Number of permutations	2000
Mean permuted coefficient	0.0004
SD permuted coefficient	0.0118
Permutation p-value (two-sided)	0.038
Actual / SD(permuted)	2.11

Notes: SCI exposure is randomly reassigned across French regions in each permutation, holding the panel structure and time fixed effects fixed. The p-value is the share of permutations where $|\hat{\beta}_{\text{perm}}| \geq |\hat{\beta}_{\text{actual}}|$. Under the null of no network effect, exposure should not predict outcomes.